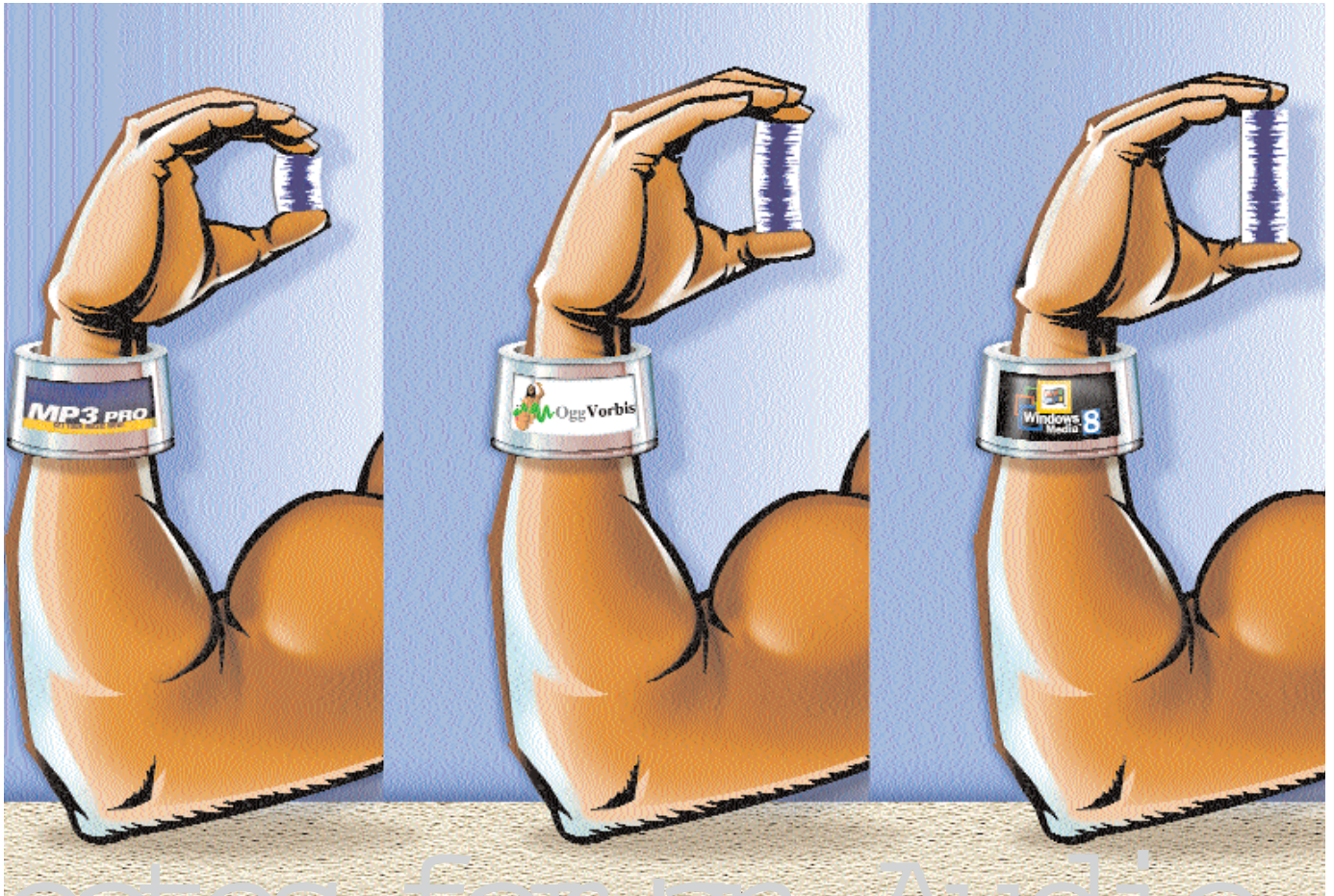




Notes for an Audiophile

A look into the world of audio formats and what's happening

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Audio formats have evolved since the early days of blips, beeps and first-generation PCM audio. It was the WAV format, which finally allowed users to experience CD-quality audio on their PCs. Though audio reproduction was remarkable, file sizes of WAV audio clips were huge. A single minute of CD-quality audio occupied approximately 10 MB of space. As an uncompressed audio format, WAV files are difficult to share and transport across computers.

The need to bring down the size of audio files was felt and development began on a format, which not only accomplished smaller audio file sizes, but also maintained as much audio integrity as possible. Fraunhofer started it all with their revolutionary MP3 format. With MP3, audio clips were cut down in size, in most cases, by a factor of 10 to 14, while maintaining audio quality to a great extent. Though audiophiles may not swear by the MP3 standard, most casual listeners

won't be able to tell the difference in audio quality between MP3 audio and CD audio.

How MP3 works

An MP3 file consists of a number of individual frames. Each frame contains a certain amount of song data and when grouped together, they make up the entire track. In order for these small segments of data to exist, a particular song must be encoded first. The bit rate chosen will directly reflect upon the output file's size and audio quality. Essentially, bit rate is a measurement (in bits per second) of how much compressed audio data flows per second. Higher bit rates yield higher quality sounding music as opposed to lower bit rates. The lower the bit rate, the less space the file occupies.

A CD contains data, which is digitally stored in an uncompressed, high-resolution manner. Music is sampled 44,100 times per second and each sample is 16 bits long. Separate samples are